AI, War & Pandemic-Resilience Stock Research — Project Charter, Plan & Roadmap

Oct 29, 2026 · @Tony

Executive summary

This project builds and maintains a ranked, evidence-based watchlist of public companies positioned for three structural forces as of late October 2026: the AI infrastructure buildout, a multi-theatre war environment (Iran, Russia-Ukraine, Indo-Pacific build-up), and the risk of a next global pandemic. The output is a categorized research report (PDF) covering 12 investment categories, refreshed quarterly against company filings.

The starting point is the prior ChatGPT research, which grew to a 109-investment PDF with Financial Strength (0–100) and Price–Fundamentals Balance scores. That report was organized by sector request rather than by theme, its doubling percentages were unvalidated, and ChatGPT itself said balance-sheet coverage was inconsistent. This project re-categorizes those holdings into 12 categories, refreshes figures with Q3 2026 results, and adds a consistent scoring model.

Market backdrop at launch: the S&P 500 closed at 7,744 on Oct 28, 2026, near records, with the 10-year Treasury above 5%, the Fed funds rate at 3.75–4.00% after a September hike, oil above $100, and US midterm elections on Nov 3. AI fundamentals remain very strong (Nvidia guided $108B for its October quarter; TSMC guided Q4 to about +58% y/y), but hyperscaler free cash flow is under strain and defense stocks have de-rated despite record backlogs.

Important: this is research and education, not personalized investment advice. Claude is not a licensed financial advisor. Scores are judgment-based screens, not return forecasts. Verify prices and filings before acting, size positions to your own risk tolerance, and consider a licensed advisor.

Project charter

Item	Definition
Project name	AI, War & Pandemic-Resilience Stock Research (P-AIWP)

Item	Definition
Sponsor / decision owner	You (the investor)
Research lead	Claude (analysis, data refresh, report production)
Start date	Oct 29, 2026
Baseline report	expanded_stock_research_109_investments_2026-10-28.pdf (ChatGPT)
Review cadence	Monthly watchlist check; full refresh after each earnings season

Purpose. Identify the strongest listed companies in 12 categories that benefit from AI adoption, sustained high defense spending, and pandemic preparedness, and keep that view current as evidence changes.

Objectives

1. Define 12 investment categories without overlap: 10 covering the AI, war and pandemic themes, plus 11 Solar & Renewables and 12 Fintech & Consumer.
2. Select 3–5 leaders per category using one transparent 100-point scoring model.
3. Re-categorize all 109 investments from the baseline PDF into the new structure, keeping prior Financial Strength and Price–Fundamentals Balance scores where still valid.
4. Refresh every top pick with its latest quarterly results (Q3 2026 season) and cite the filing.
5. Publish an updated PDF report plus this charter, plan and roadmap.

In scope: US-listed equities (NYSE, Nasdaq) and major foreign leaders available as ADRs or on home exchanges; theme ETFs as diversification alternatives; fundamentals, valuation, catalysts and risks.

Out of scope: personalized allocation advice, options and leverage strategies, trade execution, tax planning, private companies (e.g., pre-IPO AI labs).

Deliverables

#	Deliverable	Format	Status
D1	Project charter, plan and roadmap	This doc	Done

#	Deliverable	Format	Status
D2	12-category framework + scoring models	This doc, section 3	Done
D3	Updated categorized research report	PDF	Done (v2.1 with trained ML model, 24 pages)
D4	Quarterly refresh of scores and catalysts	PDF revision	Recurring

Success criteria

- Every top pick has a cited source dated within the last 120 days.
- Each of the 109 baseline holdings is assigned to exactly one of the 12 categories (or the fund group).
- Scores are reproducible: another reader applying the rubric lands within ±10 points.
- The report states the bear case for every top pick, not only the bull case.

Constraints: public information only; prices move daily, so valuation figures are as of the report date; some non-US defense leaders (Rheinmetall, BAE) are not on US exchanges.

Assumptions: AI capex stays elevated through 2027 (hyperscalers guide group capex above $1 trillion next year); NATO's 5%-of-GDP-by-2035 target holds; H5N1 remains a low-probability, high-impact human pandemic risk.

Investment framework: 12 categories and scoring model

The project's main goal is to find stocks with a high chance of at least doubling by October 2029, which needs about 26% compound annual price gain. All 117 holdings (100 from v1, 8 v2 additions, 9 funds) sit in 12 categories (1–10 for the AI, war and pandemic themes, 11 Solar & Renewables, 12 Fintech & Consumer) plus a fund group, and each gets a model probability of a 2x outcome.

#	Category	Theme	Holdings	Best doubling candidate (final P 2x)
1	AI semiconductors: compute & foundry	AI	NVDA, AVGO, TSM, AMD, QCOM, AMBA	NVDA (36%)

#	Category	Theme	Holdings	Best doubling candidate (final P 2x)
2	AI memory & semiconductor equipment	AI	MU, SK hynix, ASML, KLAC, TER	SK hynix (33%)
3	Hyperscalers & AI platforms	AI	GOOGL, AMZN, META, MSFT, APP	APP (46%)
4	Physical AI & robotics	AI	ISRG, SYK, SYM, TSLA, FANUC, Keyence + 10 more	SYM (27%)
5	AI power & data-center infrastructure	AI + energy security	VRT, GEV, ETN, CEG, NEE, AMSC	VRT (37%)
6	AI software & cybersecurity	AI + cyber war	PLTR, CRWD, PANW, ZS	PLTR (28%)
7	US defense primes & defense electronics	War	LMT, RTX, NOC, GD, LHX, CACI, DRS + 3 more	CW (22%)
8	Defense tech, drones & allied rearmament	War	AVAV, KTOS, RCAT, ONDS, Rheinmetall, BAE, Elbit	ONDS (27%)
9	Strategic resources: nuclear fuel & critical minerals	War + AI power	CCJ, BWXT, MP, Lynas, FCX + 7 more	Lynas (29%)
10	Biotech, pandemic preparedness & AI drug discovery	Pandemic	REGN, TMO, ABT, GILD, MRNA, TEM + 7 more	REGN (23%)
11	Solar & renewables	Energy transition	NXT, FSLR, ENPH, ARRY, ENLT, TOYO + 11 more	NXT (34%)
12	Fintech & consumer (outside the three themes)	Outside themes	NU, SOFI, TBBK, MELI, PDD, CELH, LULU	NU (40%)

Doubling-prediction model (ensemble)

Component	How it works	Weight in rules model
Scenario simulation	Bear / base / bull 3-year price multiples (v1 scenarios, 25/50/25), each a lognormal with 25% volatility; adjusted for Q3 results after v1's cutoff	50%
Fundamentals factor model	Doubling Potential Score: growth 30, valuation headroom 20, financial strength 20, theme tailwind 15, downside risk 15; mapped to probability with a capped logistic	50%
Tiers	Applied to the final score (50% rules + 50% trained ML model). A: 30%+, FS 70+, ML loss risk 15% or less · B: 30%+ otherwise · C: 25–29% · D: 18–24% · E: under 18%	—

Update (v2.1): trained machine-learning model. The rules model is now blended 50/50 with a model trained on 66,010 historical cases (690 stocks, 1967–2023 start dates, Yahoo Finance monthly prices). In walk-forward tests on unseen years, its top 10% of picks doubled 31–34% of the time versus an 18–22% base rate (AUC 0.58–0.63); a companion model predicts 50%+ losses (AUC 0.75). Final leaders: APP 46%, NU 40%, VRT 37%, NVDA 36%, AVGO 35%.

Six names reach Tier A on the final score (30%+ chance of doubling, Financial Strength 70+, loss risk 15% or less): NU 40% (Category 12), NVDA 36%, AVGO 35%, TSM 32%, PDD 32% (Category 12) and GOOGL 31%. The highest scores with higher risk are APP 46%, VRT 37% and NXT 34% (Category 11). Read the ranking as more reliable than the absolute percentages.

Project plan

The work runs in five phases; phases 1–3 finish with the first report, and phases 4–5 repeat every quarter.

1. **Discovery and baseline (week 1).** Import the 109-investment PDF; extract each ticker, prior scores and notes; flag stale data.
   - Gate G1: every baseline row parsed and assigned an ID.
2. **Framework and categorization (week 1).** Fix the 12 categories and the scoring models; map each baseline holding to one category or the fund group.
   - Gate G2: no holding in two categories; legacy list explained.
3. **Deep research and scoring (weeks 1–2).** Pull Q2/Q3 2026 results from SEC filings and company releases; score top candidates; write bull, base and bear cases.
   - Gate G3: every top pick has a filing citation under 120 days old.
4. **Report production (week 2).** Build the categorized PDF: summary tables per category, company pages, risk matrix, appendix of all holdings.
   - Gate G4: page-by-page visual check; numbers re-checked against sources.
5. **Monitoring and refresh (ongoing).** Track catalysts, re-score after each earnings season, log changes.
   - Gate G5: change log published with each revision.

Roles

Activity	You (sponsor)	Claude (research lead)
Approve categories and rubric	Decide	Propose
Data gathering and scoring	Review	Do
Position sizing and trades	Decide and do	Not involved
Quarterly refresh trigger	Request	Do
Final sign-off on report	Approve	Deliver

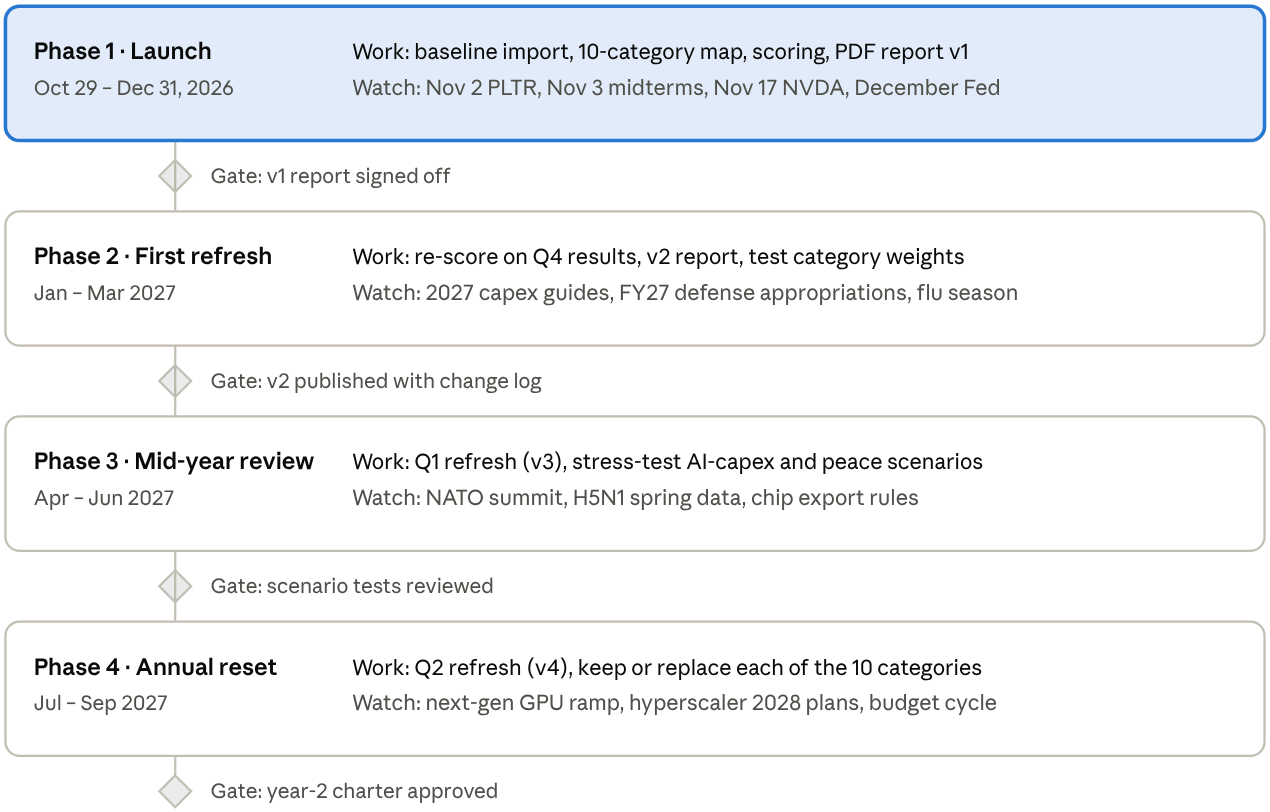
Open items for you

- [ ] Attach the baseline PDF (ChatGPT's share page blocks file access) so all 109 rows carry over exactly.
- [ ] Confirm whether non-US listings (Rheinmetall, BAE) are investable in your brokerage account.
- [ ] Choose a review day each month.

Roadmap

The first report ships in Phase 1 (now through Dec 31, 2026); each later quarter re-scores the list after earnings season and closes with a published revision.

Four quarterly phases, each closed by a report gate



12-month roadmap · 4 phases, 4 gates

Phase 1 is highlighted because it is underway; dates for Phases 2–4 move with each company's reporting calendar.

Risk register

The biggest risk to this watchlist is concentration: categories 1–6 all depend on the same AI capex cycle, so a capex pause would hit half the list at once.

Risk	Likelihood	Impact	Hits categories	Early signal	Mitigation
AI capex slowdown or "payback" sell-off	Medium	High	1–6	Alphabet FCF negative $9.1B in Q3; Microsoft trimmed capex to $170B	Cap AI categories at a set share of the watchlist; favor names with backlog (GEV, VRT) and cash (MSFT)
Higher-for-longer rates	Medium	High	1, 3, 4, 5	10-yr above 5%; Fed hiked in Sept, December hike priced	Weight the price–fundamentals factor; avoid 100x+ P/E names
Memory price reversal	Medium	High	2	DRAM contract prices; SK hynix supply commentary	Treat MU's ~6x forward P/E as cyclical, not cheap
Taiwan Strait crisis	Low	Very high	1, 2	PLA exercises, export-control news	TSM is single-point risk; pair with ASML / US fabs
Defense budget cuts after Nov 3 midterms	Medium	Medium	7, 8	House control result; FY27 appropriations	Prefer backlog-heavy primes (LMT $292B, RTX $340B) over program-dependent names
Ceasefire or peace deals (Iran, Ukraine)	Low–Medium	Medium	7, 8, 9	Negotiation headlines	Backlogs take years to burn; restocking continues after wars
Pandemic never arrives	High	Low–Medium	10	WHO still rates H5N1 public risk low	Choose companies that grow without a pandemic (TMO, ABT, GILD)
Single-binary biotech events	Medium	High	10	Trial readouts, FDA actions	Size MRNA-type names small; it rose 600%+ in 2026 on one trial
Export controls on chips	Medium	Medium	1, 2	Commerce rules	Nvidia already guides with zero China data-center revenue
Data and AI-tool error in this research	Medium	Medium	All	Figures that conflict across sources	Cite primary filings; re-check before any decision

Research program

A standing research track looks for a model that beats the current one (EXP-000) at ranking stocks by their chance of doubling in 3 years and flagging likely halvings. The full method is in research/RESEARCH_FRAMEWORK.md; every experiment is logged in `research/experiments.csv`.

Factor	Path A: original design	Path B: combine existing strengths
1. Define success	Same metrics for both: walk-forward AUC, top-decile hit rate, calibration, loss-model AUC, top-decile basket vs S&P 500	Same, plus limits on data and licensing of borrowed models
2. Build a baseline	EXP-000 plus momentum, low-volatility and random rankings	Benchmark each existing model family on our task; record pros and cons
3. Hypothesis	A new mechanism aimed at a specific failure (e.g. a market-regime gate)	A combination that covers individual weaknesses (e.g. price ML + SEC fundamentals)
4. One change at a time	Ablation of each mechanism; point-in-time data, 36-month embargo, 3+ seeds	Add or remove each borrowed component; compare with source models and a simple average
5. Refine and scale	Keep what holds across eras; pass an untouched holdout	Drop components that add nothing; scale survivors

Promotion rule: a new model must improve AUC by at least 0.02 in 2 of 3 eras (no era worse by more than 0.01) and raise the top-decile hit rate, without worse calibration or loss-model AUC. **Gates:** R1 plan approved, R2 validation passed, R3 holdout passed, R4 sponsor adopts it as the new baseline.

First cycle backlog: R-01 SEC point-in-time fundamentals (B) · R-02 market-regime gate (A) · R-03 time-to-double survival model (A) · R-04 learned weights for the rules model (B) · R-05 survivorship-bias study (data search) · R-06 live scoring of the Oct 2026 predictions (running). The framework is reusable for Forex and options by changing target, horizon and costs.

Governance, review rules and KPIs

A holding moves only on evidence written into the change log, never on price action alone.

Review rules

- **Upgrade** to top pick: score reaches 70+ and the latest quarter beat and raised guidance.
- **Downgrade** to watch: score falls below 60, guidance cut, or the thesis-defining contract or trial fails.
- **Remove**: fraud, restatement, going-concern warning, or the company leaves the theme.
- **Theme check** each quarter: if a theme's driver weakens (e.g., hyperscaler capex guidance falls two quarters in a row), cut that category's weight in the report.
- **Diversification guide**: keep any single category to a stated cap of the watchlist so no one theme dominates; ETFs (e.g., ITA for US defense, CIBR-type cyber funds) are listed as lower-risk alternatives.

Delivery workflow (standing rule from the sponsor, Oct 29, 2026). After completing any request in this project, Claude finishes in this order:

1. **Local GitHub folder first:** save all relevant information and data to `~/Documents/GitHub/p12-best-stocks-investment` (reports, docs, data, scripts, results, chat notes), updating `README.md`, `CHANGELOG.md` and the checksum manifest.
2. **Website section second:** update the related part of the live page source, `website/project-12.html`, in the same folder.
3. **GitHub and website, automatic:** the sponsor commits and pushes the folder to hamidf2024/p12-best-stocks-investment with GitHub Desktop; the workflow `sync-website.yml` then publishes the page to hamidf2024.github.io/stocks-investment/project-12.html.

A request is not complete until all three steps are done, or a step that could not be done is reported with the reason.

KPIs for the project

KPI	Target
Top picks with filing cited within 120 days	100%

KPI	Target
Baseline holdings categorized	109 of 109
Report refresh after earnings season	Within 3 weeks of the last top-pick report
Bear case written per top pick	100%
Watchlist vs S&P 500 (tracking only, not a promise)	Reported each quarter

Sources

- Baseline ChatGPT conversation
- NVIDIA Q2 FY2027 results (SEC 8-K)
- Nvidia Q2 takeaways and $108B guide (CNBC)
- Broadcom Q3 FY2026 results (SEC 8-K)
- TSMC Q3 2026 results (SEC 6-K)
- ASML Q3 2026 presentation (SEC 6-K)
- Micron valuation and FQ4 2026 results
- Alphabet Q3 2026 results (SEC 8-K)
- Hyperscaler capex and FCF, Oct 29 2026 (CNBC)
- Palantir Q2 2026 results and guidance (SEC 8-K)
- GE Vernova Q3 2026 results (SEC 8-K)
- Vertiv Q3 2026 results (SEC 8-K)
- CrowdStrike Q2 FY2027 results
- RTX Q3 2026 results (SEC 8-K)
- Defense earnings week, Oct 21 2026 (Nasdaq)
- Lockheed guidance raise (24/7 Wall St.)
- Rearmament cash flow and Rheinmetall (MarketScreener)
- Drone stocks year-to-date (24/7 Wall St.)
- Cameco quote page (StockAnalysis)
- Thermo Fisher Q3 2026 results (Nasdaq)
- Abbott Q3 2026 results

- Moderna 2026 rally (Motley Fool)
- WHO avian influenza weekly update, Oct 9 2026
- S&P 500 close and Fed decision, Oct 28 2026
- Geopolitical risk assessment, October 2026